Senior Game Rental Database System Report

1. Introduction

This project involves the design and implementation of a relational database system intended to support the operations of a Senior Game Rental platform, which focuses on enhancing cognitive and social engagement among elderly users. The system is designed to manage various types of data, including user information, game inventories, booking records, payments, feedback, and cognitive performance metrics.

The primary goal of this database is to provide a structured and efficient way to store and analyze data related to senior participation in game-based activities. By organizing this data effectively, the system enables stakeholders to monitor usage patterns, evaluate user satisfaction, and assess cognitive development over time.

2. Database Design

I developed the database using MySQL Workbench. The database created for this project is named senior_game_rental.

The design follows relational database principles where data is organized into multiple tables connected using primary and foreign keys.

Main tables include:

- users
- students
- seniors
- games
- consoles
- bookings
- booking_games
- payments
- session_feedback
- cognitive_scores

3. Data Population

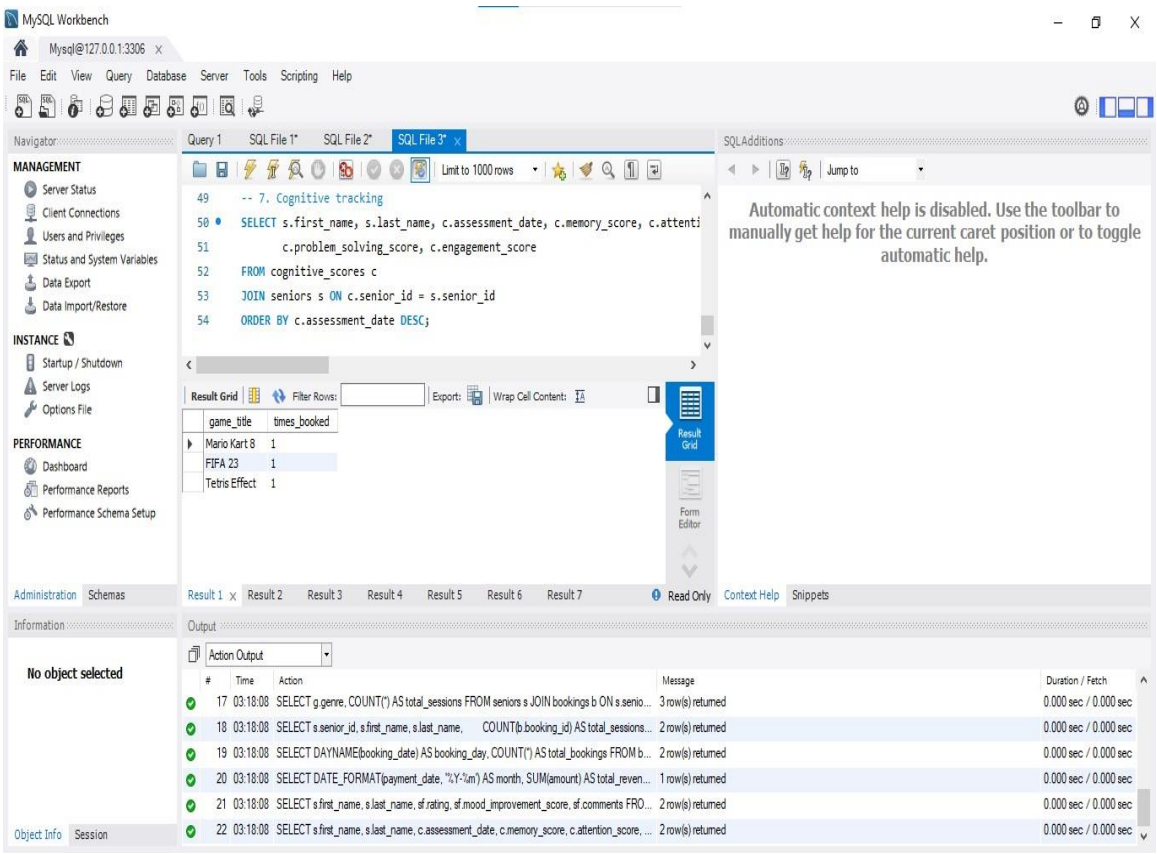
Sample data was inserted into all tables using SQL INSERT statements to simulate real-world scenarios such as bookings, payments, feedback, and cognitive tracking.

4. Query Implementation and Results

4.1 Most Popular Games

This query identifies the most frequently rented games, helping understand user preferences.

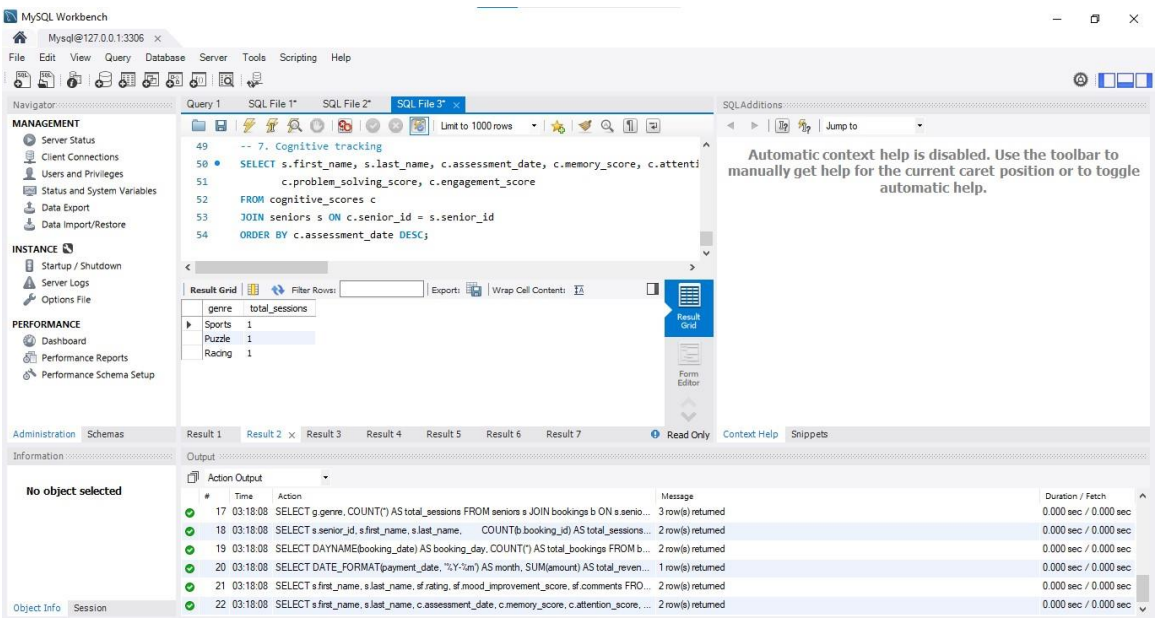
Screenshot – Result 1



4.2 Game Usage by Age Group (75+)

This query analyzes game usage among seniors aged 75 and above.

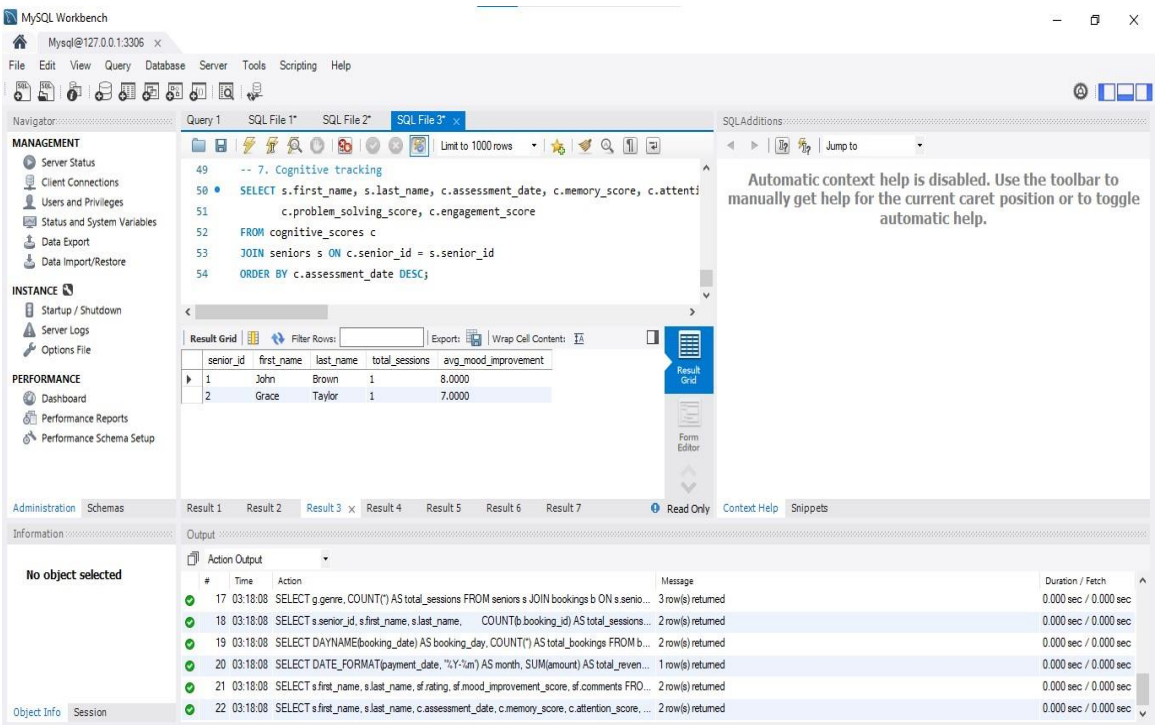
Screenshot – Result 2



4.3 Session Frequency and Mood Improvement

This query evaluates how session participation affects mood improvement.

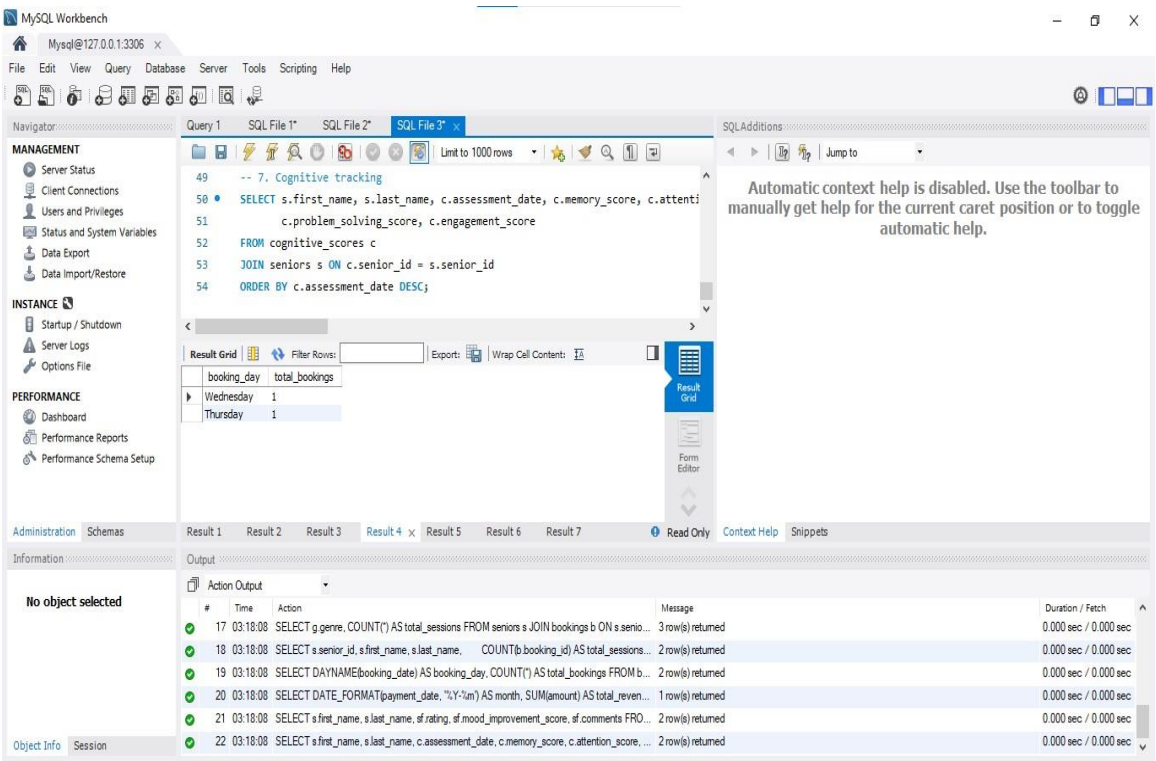
Screenshot – Result 3



4.4 Peak Booking Days

This query identifies the days with the highest booking activity.

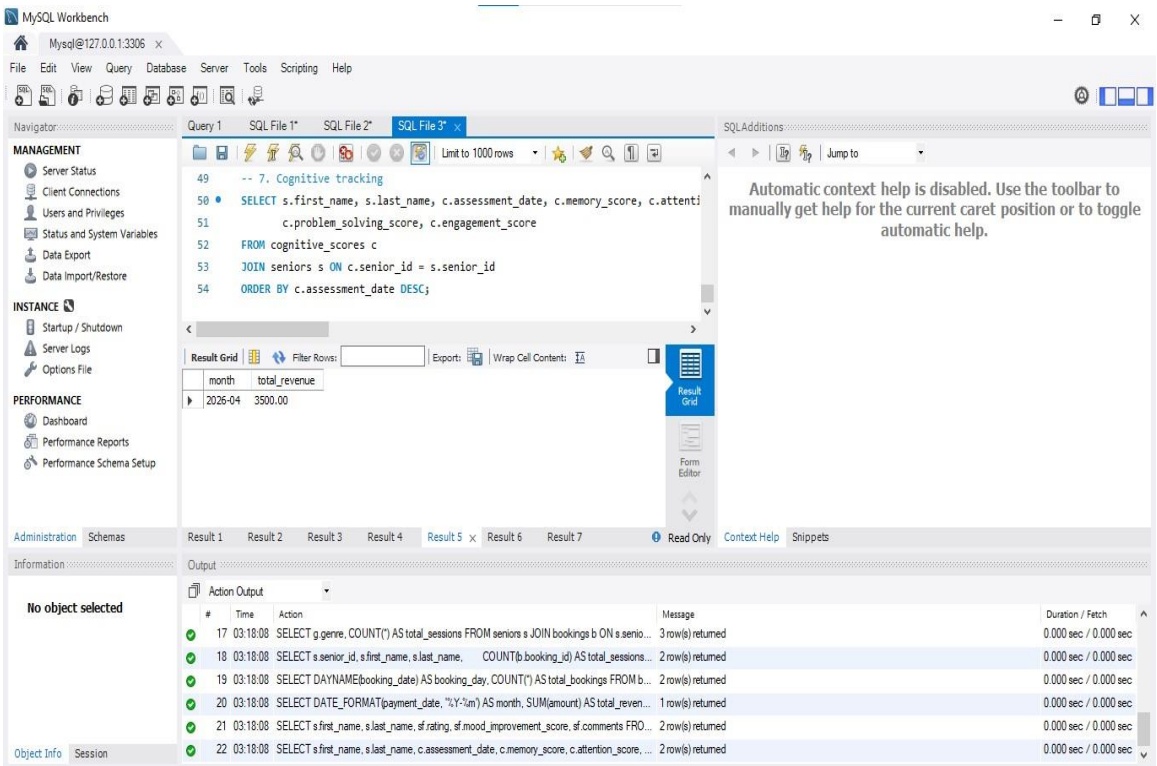
Screenshot – Result 4



4.5 Revenue Analysis

This query calculates total revenue trends over time.

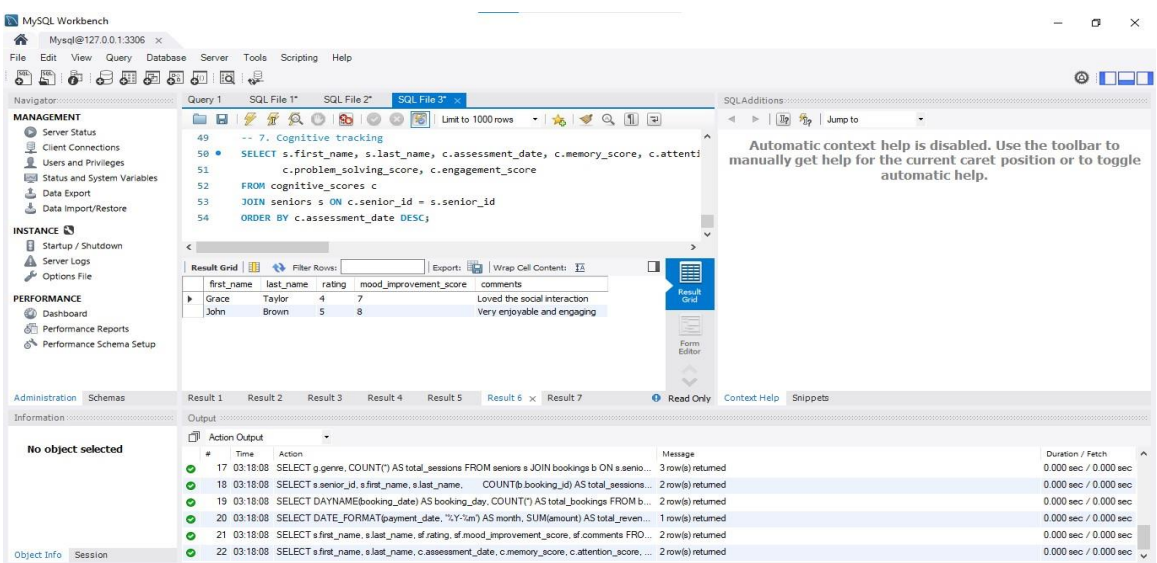
Screenshot – Result 5



4.6 Feedback and Satisfaction

This query analyzes feedback and satisfaction levels among the seniors.

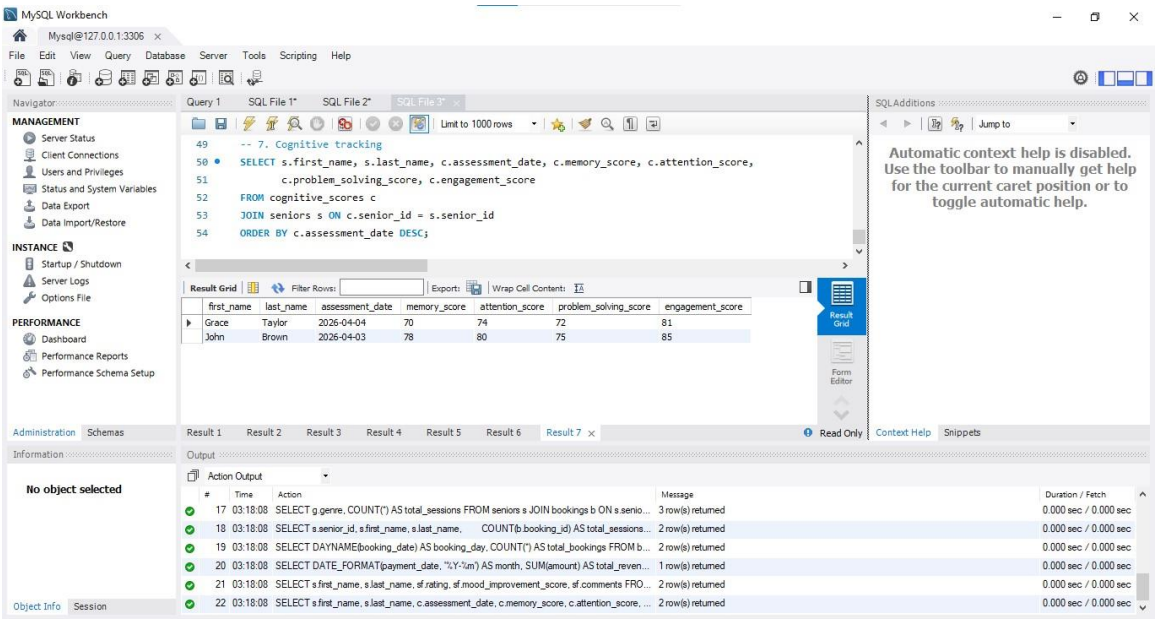
Screenshot – Result 6



4.7 Cognitive Performance Tracking

This query tracks cognitive development metrics such as memory and attention.

Screenshot – Result 7



5. Discussion

The database demonstrates how structured data can be used to analyze user behavior, financial performance, and cognitive development. It supports data-driven decision-making and service improvement.

6. Conclusion

The system successfully manages and analyzes data for a senior game rental platform. It demonstrates practical database design and real-world application potential, especially in supporting elderly cognitive engagement.